

Chap 7 Worked problems :

$$W_Q = W_K = W_V = W_H = D$$

1) MLP: 3LDF

$$W_Q = W_K = W_V = D \times H$$

$$\text{Attn} = 2LDH(N+K)$$

Vocab embeddings: VD

$$\therefore \text{Total} = 3LDF + 2LDH(N+K) + VD$$

$$= 12.885B + 5.369B + 0.132B \approx 18.4B //$$

KV cache size per token in int8 ! , K=heads, H=per head

For every token, the cache at one layer = $2KH$

$$\therefore \text{Total} = 2LKH = 262,144 \text{ bytes} // \approx 262 \text{ KB} //$$

2) Total mem for devices = $16 \times 4 \times 4 = 256 \text{ GB}$

$$\text{Model weights} = 18.4 \text{ GB}$$

$$\text{KV Cache size} = 262 \text{ KB} \times 128,000 = 33.556 \text{ GB per seq.}$$

$$\therefore B_{\max} = \left\lfloor \frac{256 - 18.4}{33.55} \right\rfloor = 7 //$$

$$\text{If } K=1, \text{ then KV cache size} \times \frac{1}{8} \Rightarrow B_{\max} \times 8 = 56 //$$

$$3) \text{ time} = \frac{\text{bytes loaded}}{BW}$$

$$\text{Each chip BW} = 8.2 \times 10^9 \text{ bytes/s} \Rightarrow \text{Total BW} = 16 \times 8.2 \times 10^9 = 1.312 \times 10^{12}$$

$$\text{bytes} = 18.4 \times 10^9 \text{ bytes}$$

$$\therefore \text{time} = \frac{18.4 \times 10^9}{1.312 \times 10^{12}} \approx 1.4 \text{ ms}$$

4) a) What does ICI look like:

2D physical dim 4-way along rows, 4-way along cols.

b) Tensor parallelism roofline: Where ICI slower than weight loading

(Decode) $T_{ICI} = \frac{2SD}{W_{ICI}}$

$$T_{HBM} = \frac{\text{bytes}}{p \cdot W_{HBM}} = \frac{2DF}{p \cdot W_{HBM}}$$

$$T_{ICI} = T_{HBM} \Rightarrow \frac{2SD}{W_{ICI}} = \frac{2DF}{p \cdot W_{HBM}} \Rightarrow p = \frac{W_{ICI}}{W_{HBM}} \left(\frac{F}{B} \right) = \frac{F}{B} \approx \frac{F}{8B}$$

Using prev $F=16384$ and max $B=7$, then

$$p_{max} = \frac{16384}{8 \cdot 7} \approx 293 \gg 16 \Rightarrow \text{very much below bandwidth roofline and weight loading is still the bottleneck.}$$

(Prefill) $p_{prefill} = \frac{F}{2200} = \frac{16384}{2200} \approx 7.4 \Rightarrow \text{around 8-way tensor parallelism.}$

c) How to shard KV cache: $[2, B, S, K, 4]$

1. KV-head dim. K , K limited to 8-way

2. B

3. S

We can do 4×4 or 8×2 , but 4×4 should have better balance overall in case of comm.

- Natural 4x4 sharding = [2, B, S, K, Z, H] $\bar{Y} = Z = 4$

$$T_{\text{step}} = T_{\text{ker load}} + \max(T_{\text{param head}}, T_{\text{compute}})$$

- batches

$$T_{\text{ker load}} = \frac{33.55615 \times 7}{1.312 \times 10^5} \approx 17.9 \text{ ms}$$

$$T_{\text{params}} = 1.4 \text{ ms} \quad (0.3)$$

$$T_{\text{compute}} = \frac{25P}{16C} = 0.001 \text{ ms} \ll T_{\text{params}}$$

$$\therefore T_{\text{step}} = 17.9 + 1.4 = 19.3 \text{ ms}$$

5) a) Total = 3ELDF + 2LDHGN4K + DV $\approx 212B$

Activated = 3KLDF + 2LDHGN4K + DV $\approx 31.2B$

b) To be FLDB bound, $T_{\text{math}} = T_{\text{HBM}}$

$$T_{\text{HBM}} \propto \frac{EM \cdot \text{bytes/param}}{WHBM}$$

$$T_{\text{math}} = \frac{28 \text{ km}}{C}$$

For bf16 inf, TRV50 Bcm ≈ 240 . MOE increases this by $\frac{F}{K} = 8 \Rightarrow$
 $B_{\text{cm}} = 240 \times 8 = 1920 \text{ tokens.}$

c) MOE changes MLP, not attn, $\therefore 26.2 \text{ KB/token still}$

d) FLOPs $\approx 2 P_{\text{activated}} T = 62.4 \times 10^9 \times T$